# Cointegration Summary — Week 7 Part 1

Oct 10, 2026 · @Keehong Rhee

## At a glance

Cointegration is the property of two series that each wander without bound, yet stay around a fixed level when tied together in one particular combination. That combination is the long-run equilibrium, and it becomes the anchor for pairs trading. Part 1 comes down to three points.

1. **Cointegration is a stationary combination of wandering series.** For two I(1) series X and Y, if Y − βX is stationary, the two are cointegrated. They share one common trend, and the spread is the deviation from the long-run equilibrium.
2. **It separates real relations from spurious regression.** A levels regression is genuine only when its residuals are stationary. A stationary spread is the certificate, and testing for it is the subject of Part 2.
3. **The spread is a tradable anchor.** Prices rarely return to a mean, but a cointegrated spread does. It is a rare signal with an economic basis, and the foundation of pairs trading.

## 1. Starting point: the long-run relation that differencing threw away

The Week 6 VAR required stationary series, so trending prices were differenced before going in. But differencing erases the long-run relation contained in the price **levels**. Cointegration is exactly the concept that recovers that relation.

### Basic idea: I(1) and I(0)

The order of integration means "how many times a series must be differenced to become stationary". The simplest example is a random walk.

$$y_t = y_{t-1} + \varepsilon_t \;\Rightarrow\; y_t \sim I(1), \qquad \Delta y_t = y_t - y_{t-1} = \varepsilon_t \sim I(0)$$

- **I(1)** (integrated of order 1): as in a random walk, shocks keep piling up and there is no mean to return to. Price levels such as stock prices and exchange rates are typical examples. It is the same thing as the unit-root process from Week 2.
- **I(0)** (stationary): moves up and down around a fixed mean and variance, and comes back after it strays. Returns (the differences of prices) are the typical example.
- Differencing once turns I(1) into I(0), but at the cost of throwing away the information in the levels.

### The problem carried over from Week 6

1. **A VAR needed stationary series.** Trending prices were differenced first, and in the process the long-run relation could be lost.

2. **Some pairs share a long-run equilibrium.** Two non-stationary prices can move together forever for economic reasons. Differencing both cuts that tie.
3. **Cointegration models that tie directly.** It captures the stable relation in levels, and error correction (Part 3) explains how deviations are pulled back. This is the engine of pairs trading.

### Episode: the problem Granger solved twice

- **1974: exposing the trap.** Granger, with Newbold, exposed the trap of spurious regression (Week 2). Regressing two unrelated random walks on each other gives a high $R^2$, but the relation is a statistical illusion. The result became a warning to a generation of researchers: "do not regress levels on levels".
- **The open question.** If levels regressions are usually dangerous, is there never a case where the relation between levels is **real**?
- **1987: offering the solution.** Engle and Granger defined cointegration and its partner, the error-correction model. Some I(1) series do share a genuine long-run equilibrium, and if a particular combination is stationary, the levels regression is not spurious. This work led to the 2003 Nobel Prize in Economics.

There are two lessons. Not every levels regression is a trap; if the series are cointegrated, the relation is real. And that stationary combination is an economic anchor — exactly the anchor that pairs trading exploits.

## 2. Definition of cointegration

Two series are cointegrated if each is I(1) but some fixed linear combination of them is I(0). The definition fits on one line.

$$X_t \sim I(1), \quad Y_t \sim I(1), \qquad z_t = Y_t - \beta X_t \sim I(0)$$

- X and Y are each non-stationary (I(1)). Neither has a mean to return to.
- Yet the spread $z = Y - \beta X$ is stationary (I(0)). This is the definition of cointegration.
- This property holds for **one particular β only**. With any other β, the combination still wanders. Finding that β is the heart of cointegration analysis.

### Cointegrating vector and hedge ratio

Writing the weights in the definition as a vector gives the cointegrating vector.

$$z_t = \underbrace{\begin{pmatrix} 1 & -\beta \end{pmatrix}}_{\text{cointegrating vector}} \begin{pmatrix} Y_t \\ X_t \end{pmatrix} = Y_t - \beta X_t$$

- $(1, -\beta)$ is the set of weights that makes the combination stationary. Setting the first element to 1 is called normalisation.

- z is the equilibrium error, that is, the spread we watch.
- In the language of trading, β is the **hedge ratio**. It tells how many units of X are needed to hedge one unit of Y, and it defines a market-neutral spread position.

## 3. Why it is special: the bookkeeping of integration orders and the common trend

### Normally impossible

Mixing two I(1) series generally gives I(1) again, because the wandering components add up rather than cancel.

$$X_t \sim I(1),\ Y_t \sim I(1) \implies aX_t + bY_t \sim I(1) \quad \text{(in general)}$$

So a stationary combination ought to be impossible. The only exception is when the two series share **the same stochastic trend**. Then the right β subtracts one trend from the other, leaving only a stationary remainder.

### Seen through the common trend

Cointegrated series can be written as "one shared random walk + their own stationary noise". Call the common trend τ.

$$\begin{aligned} \tau_t &= \tau_{t-1} + \eta_t \qquad \text{(common stochastic trend)} \\ X_t &= \tau_t + u_t, \qquad Y_t = \beta\,\tau_t + v_t \\ z_t &= Y_t - \beta X_t = v_t - \beta\,u_t \ \sim\ I(0) \end{aligned}$$

- First line: τ is a random walk. All the non-stationarity of both series comes from here.
- Second line: X is τ and Y is βτ, each plus its own stationary noise u, v.
- Third line: forming Y − βX cancels βτ exactly, leaving only stationary noise. That is why the spread is I(0).

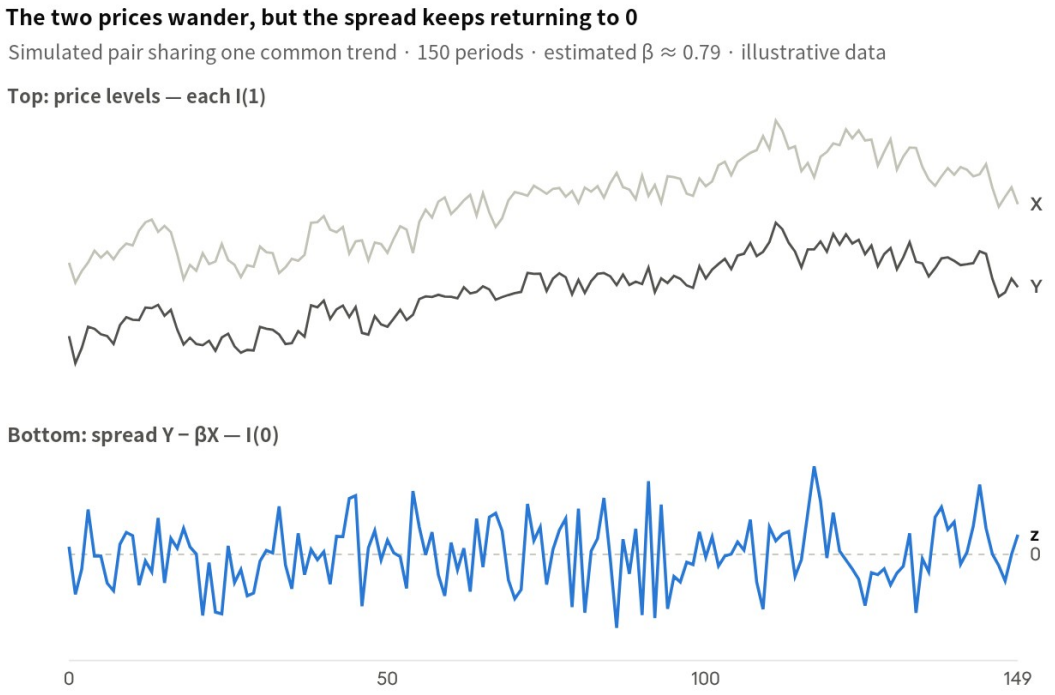
From this angle, cointegration means "fewer trends than series". If two series move on one trend, they are cointegrated; if each has its own trend, no combination can remove them.

| Case | Trend structure | Spread Y − βX |
|---|---|---|
| Cointegrated | The two series share one trend | I(0) at the right β — returns to its mean |
| Not cointegrated | Each has its own independent trend | I(1) at any β — keeps wandering |

## 4. Intuition: a drunk and her dog

A drunk walks aimlessly, and her dog wanders aimlessly too. Both are random walks. But if a leash ties them together, **the distance between them** is stationary.

| Two random walks | Leash = cointegration |
|---|---|
| The person's position wanders unpredictably | Because of the leash, the two cannot drift apart |
| So does the dog's position | The gap between them mean-reverts around the leash length |
| Neither returns to a set place on its own | That bounded gap is the stationary spread |




leash.csv · common-trend simulation, 150 periods (illustrative data)

Neither price in the top panel returns to a mean, yet they always move together. The spread in the bottom panel keeps going back and forth around 0. As the lecture puts it, the two are tethered, not tied down.

## 5. The cure for spurious regression

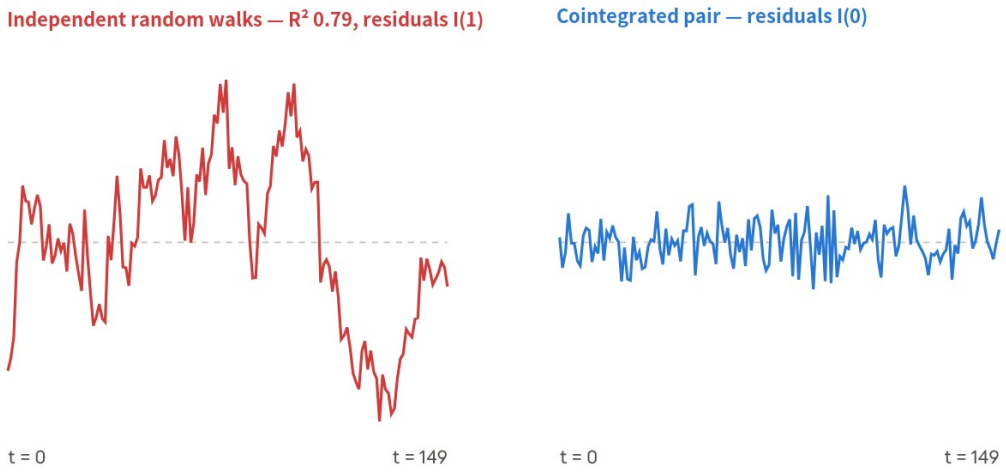
The warning in Week 2 was "do not regress trending series on each other". Cointegration is the exact exception to that warning. The residuals of the regression decide.

$$Y_t = \alpha + \beta X_t + e_t : \quad \begin{cases} e_t \sim I(1) & \Rightarrow \text{ spurious regression} \\ e_t \sim I(0) & \Rightarrow \text{ cointegration (genuine)} \end{cases}$$

| | Spurious regression (Week 2) | Cointegrated |
|---|---|---|
| $R^2$ | High but meaningless | Meaningful |
| Residuals | The residuals themselves wander — I(1) | The residuals are stationary — I(0) |
| Relationship | A mirage | Certificate of a real relationship |

**Spurious-regression residuals wander; cointegrating ones return to 0**

Residuals of the levels regression Y = α + βX · same vertical axis · 150 periods · illustrative data




resid.csv · levels-regression residuals, 150 periods (illustrative data)

A pair of independent random walks reaches an $R^2$ of 0.79, yet the residuals drift far to one side. The residuals of a cointegrated pair return to 0 within a narrow band. So **testing whether the residuals are stationary** is how to tell the two apart, and that is the Engle-Granger test of Part 2.

## 6. Cointegration is not correlation

Correlation measures short-run co-movement in returns; cointegration measures a long-run equilibrium in price levels. They are related but not the same concept.

$$\rho = \mathrm{Corr}(\Delta X_t, \Delta Y_t) \quad \text{vs.} \quad Y_t - \beta X_t \sim I(0)$$

| | Correlation | Cointegration |
|---|---|---|
| What it measures | Short-run co-movement in | Long-run equilibrium in levels |

| | Correlation | Cointegration |
|---|---|---|
| | returns (differences) | |
| Stability | Can be high one month and gone the next | Deviations are temporary and eventually corrected |
| Long-run relation | Says nothing about a long-run tie between levels | That tie is the definition |
| How they relate | High correlation does not imply cointegration | Cointegrated series can have weak daily correlation |

## 7. The equilibrium spread: a tradable anchor

The stationary spread z = Y − βX measures how far the pair is from its long-run equilibrium. Because the spread returns to its mean, extreme values become trading opportunities.

$$z_t > \mu_z : \text{ short } Y, \text{ long } \beta X \qquad z_t < \mu_z : \text{ long } Y, \text{ short } \beta X$$

- **z is above its mean (μ_z)**: Y is expensive relative to X. Sell Y and buy β units of X.
- **z is below its mean**: the reverse — buy Y and sell β units of X.
- **z is near its mean**: the two assets are 'fairly' priced relative to each other.

### Why finance loves cointegration

Prices are close to random walks and hard to forecast. But a cointegrated spread returns to its mean, so it gives a statistically grounded signal even where forecasting a single asset fails.

- It builds a mean-reverting quantity out of prices that do not mean-revert.
- It is a market-neutral position: buy one side, sell the other.
- It is an economic anchor, not a fitted pattern.
- Where it is used: relative-value and statistical arbitrage, index arbitrage and futures basis trading, and pairs trading in general.

### Pairs likely to be cointegrated

| Candidate | Economic force tying the two |
|---|---|
| Spot and futures on the same underlying | Arbitrage bounds the basis |
| Dual-listed shares of the same company | Claims on the same cash flows |
| Two firms in the same industry with the same drivers | Substitution and common demand shocks |

When the two drift apart, that force pulls them back together. Without such a force there is no cointegration, however high the correlation.

### When they are not cointegrated

- Every linear combination keeps wandering. The spread is I(1).
- No economic force ties the two together. Correlation may be high, but there is no equilibrium.
- Trading a 'spread' that is not cointegrated is trading a random walk. Once it widens, it may never come back. The tests in Part 2 guard against this.

### Next: concept → test → trade

| Stage | Content |
|---|---|
| Part 2 — Tests | Engle-Granger: run the regression, then test the residuals for stationarity. Johansen: test many series at once and read the cointegrating rank (how many equilibria there are) |
| Part 3 — Trading | Error correction: the force that restores equilibrium. VECM and adjustment speed. Pairs trading built on the cointegrated spread |

## 8. Common misconceptions and self-check

| Common misconception | Reality |
|---|---|
| High correlation means cointegration | Correlation is short-run co-movement; cointegration is a long-run tie between levels (Section 6) |
| Any two trending series can be paired | Only pairs sharing a common trend are cointegrated, and most are not (Section 3) |
| A cointegrated spread never strays far | It returns only on average, and can stray a long way first (Part 3) |

### Three questions before moving on

1. **The spread $Y - \beta X$ of two I(1) prices is stationary. What is their relationship?** They are cointegrated.
2. **How does cointegration escape the trap of spurious regression?** Because the residuals of the levels regression are stationary. Stationary residuals are the certificate of a real relationship.
3. **Can two series be highly correlated yet not cointegrated?** Yes. They are different concepts.